

Land Cover Segmentation and Classification of Satellite Imagery

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Abstract—Satellite imagery plays an important role in urban planning and environmental management, especially in tropical regions like Malaysia, where environmental diversity makes land classification rather complex. The project addresses the imbalance in the DeepEarthMY dataset, which does not have enough representation of urban images. Therefore, this project’s main contribution is enhancing DeepEarthMY with additional annotated images from the Sabah and Sarawak regions to create a more balanced dataset. Furthermore, the analysis showed significant differences in pixel and object counts in the rural and urban domains. Rural areas are dominated by Agriculture and Forest categories, while urban areas are dominated by Buildings and Roads. After that, we trained and evaluated several deep learning models, with EfficientNetV2-B2 achieving the best overall performance, making it best for land cover classification. Further comparison between the EfficientNet and VMamba model families highlighted the strengths of each, with EfficientNetV2-B2 and VMambaV2-Small performing best in their respective families. Additional experiments comparing dataset versions and split strategies revealed that random splits on the earlier version (V1) yielded higher accuracy than the manually split refined version (V2), emphasizing the importance of dataset splitting.

Index Terms—Land-cover classification, Deep Learning, Satellite Imagery, Tropical Region, Image classification model

I. INTRODUCTION

Malaysia’s tropical landscapes, which span dense forests, agricultural zones, and rapidly expanding urban centers, pose unique challenges for the classification of land cover. Identifying accurately between rural and urban areas in satellite imagery is crucial for informed decision-making in urban planning, disaster management, and environmental policy. However, high spectral variability and overlapped land use features in tropical regions often lead to misclassifications when using conventional remote sensing methods.

In recent years, deep learning-based classification has emerged as a robust solution to overcome these challenges. Using spatial and contextual information within satellite images, modern architectures such as EfficientNetV2 and VMamba have demonstrated improved performance in distinguishing broad types of land cover. In this study, we focus on binary classification of satellite images into rural and urban categories using high-resolution data. These models offer promising capabilities to capture abstract visual patterns that correlate with human-made and natural landscapes.

To support this research, we utilize the DeepEarthMY dataset, which provides satellite imagery across diverse

Malaysian regions. Despite its potential, the dataset initially suffers from underrepresentation of urban regions, particularly in East Malaysia. To address this, the dataset was enhanced with additional annotated urban images to improve class balance and support more reliable classification. The impact of this refinement, along with different data splitting strategies, is evaluated through a series of experiments comparing model performance across dataset versions.

II. LITERATURE REVIEW

A. Related Work: Models and Datasets for Land-Cover Classification

Recent advancements in deep learning have significantly enhanced the capabilities of land-cover classification in remote sensing. Early architectures like AlexNet [1] and ResNet [2] laid the groundwork for deep convolutional networks, enabling better generalization and the ability to train deeper models. Subsequent models such as ResNeXt [3] and DenseNet [4] introduced improvements in modularity, parameter efficiency, and gradient flow, though they still required considerable computational resources. More recent methods, including EfficientNetV2 [5], have achieved better performance by optimizing model scaling and training speed. Vision Transformers (ViT) [6] shifted the paradigm from convolutional to attention-based mechanisms, excelling particularly on large datasets. Hybrid and advanced architectures such as MaxViT [7] combine convolution and self-attention to capture both local and global dependencies. In contrast, VMamba [8], based on state space models rather than transformers, offers a linear-complexity alternative for long-sequence modeling. MambaVision [9] builds on this by integrating Mamba modules into visual backbones, enabling efficient large-scale image analysis without relying on attention mechanisms.

A wide variety of benchmark datasets have been developed to support model evaluation in land-cover tasks. Datasets such as LandCoverNet, SpaceNet, and DeepGlobe offer global coverage with varying resolutions and class diversity [10]–[12], while SAT-4 and SAT-6 focus on large-scale image volumes at low resolution [13]. More recent datasets like LoveDA and EuroSat include higher-resolution imagery and improved class definitions for urban-rural segmentation [14], [15]. The DeepEarthMY dataset [16], used in this study, provides equatorial satellite imagery tailored to the Malaysian landscape but suffers from underrepresentation of urban regions and East Malaysia. Enhancing this dataset to improve class balance and

geographical coverage is essential for achieving more robust classification in tropical contexts, where traditional models often underperform due to overlapping spectral signatures and complex land-use patterns.

III. METHODOLOGY

A. Image Annotation

The original DeepEarthMY dataset is inherently imbalanced, with a significantly higher number of rural images (2,605) compared to urban images (1,402). Additionally, rural samples tend to dominate land-cover classes such as agriculture and forest, while urban samples are more concentrated in buildings and roads. This imbalance in both image distribution and pixel-level class representation can bias model training and hinder generalization, particularly in accurately classifying underrepresented urban features.

Therefore, additional annotated images were added to correct any imbalances and improve the representation of urban images. We specifically enhanced the amount of urban images to provide a more balanced distribution between rural and urban classes.

The additional images include 300 images from Sabah, covering densely built-up areas in Kota Kinabalu, coastal landscapes in Sandakan, and plantation-adjacent zones in Tawau. Another 327 images were sourced from Sarawak, capturing diverse urban environments in Kuching, industrial sites in Miri. As highlighted in Table I, high-density regions—such as Kota Kinabalu with over 5,200 people per km²—were prioritized to improve urban representation. With these additions, the updated DeepEarthMY v2.0 now consists of 4,634 images: 1,965 urban and 2,669 rural. This revision ensures a more balanced spatial and domain distribution, which is critical for reliable land-cover classification in equatorial settings.

TABLE I
URBAN IMAGE ANNOTATION OVERVIEW

State	District	Constituency	Population Density (people/km ²)	Images
Sabah	Kota Kinabalu	Api-Api	5251	57
	Sandakan	Elopura	5339	54
	Tawau	Kukusan	3737	62
	Sandakan	Sungai Sibuga	2045	64
Sarawak	Putatan	Petagas	2390	64
	Betong	Batu Lintang	4336	64
	Kuching	Padungan	2363	64
	Kuching	Pending	2647	40
	Miri	Pujut	2305	58
	Kuching	Kota Sentosa	1898	26
	Kuching	Samariang	2652	32
	Kuching	Satok	1446	43
Total				627

B. Dataset Splitting Strategy

The dataset was manually divided into training, validation, and testing sets using a 70:15:15 ratio to ensure both spatial independence and domain balance. To prevent data leakage and overfitting, images from the same district were not shared across splits. Additionally, the split ensured that both urban and rural images were adequately represented in all subsets to support robust evaluation across domains. As shown in Table II, the final distribution consists of 3,222 training images, 701 validation images, and 711 testing images. Urban images account for 1,344 in training, 312 in validation, and 309 in testing, while rural images comprise 1,878, 389, and 402 in each respective set. This balanced and independent split facilitates a fair assessment of model generalization in diverse geographical contexts. Moreover, Table III provides a detailed overview of how each state assembly constituency is allocated across the training, validation, and testing splits.

TABLE II
FINAL DATA SPLIT AFTER ADJUSTING FOR SPATIAL INDEPENDENCE

Domain	Train	Validation	Test
Urban	1344	312	309
Rural	1878	389	402
Total	3222	701	711

TABLE III
SPLITTING OVERVIEW IN DEEPEARTHMY 2.0 DATASET

Split	Domain	State Assemblies
Train	Urban	Kota Iskandar, Kota Lama, W.P Putrajaya, N.21 Luyang, Api-Api, Kukusan, Petagas, Seri Setia, Kampung Tunku, Batu Tiga, Kota Anggerik, Subang Jaya, Bakar Arang, Menglembu, Buntong, Batu Lintang, Padungan, Pending, Kota Sentosa, Samariang, Satok, Sura
	Rural	Tiram, Ayer Hangat, Kuah, Sungai Limau, Tokai, Titi Serong, Teluk Intan, Sungkai, Malim Nawar, Pauh, Sungai Pelek, Sungai Panjang, Permatang, Sekinchan, Tanah Rata, Permaisuri, Water
Validation	Urban	Larkin, Pandamaran, Kota Darul Aman, Pelawan, Pujut
	Rural	Bukit Kayu Hitam, Pulau Betong, Jelai, Pantai Remis, Bukit Melawati, Sungai Air Tawar, Jepak, Water
Test	Urban	Bukit Bintang, Komtar, Elopura, Sungai Sibuga, Bukit Antarabangsa
	Rural	Sungai Balang, Bukit Pinang, Slim, Chuping, Meru, Sabak, Batu Rakit, Water

C. Evaluation Metrics

The performance is evaluated by several key evaluation metrics. the metrics include: accuracy, precision, F1-Score, and recall. This helps us perform an analysis about how well or badly the model performs in predicting each land-cover type. In classification, model outcomes are separated into True Positives (TP), True Negatives (TN), False Positives (FP), and False Negatives (FN). TP and TN represent correct predictions, while FP and FN indicate misclassifications. These metrics are essential for assessing model performance. (i) Accuracy is the

proportion of correct predictions to total predictions, and is a standard metric for model evaluation.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}, \quad (1)$$

(ii) Precision measures how many of the predicted positives are actually positive. It is especially important in imbalanced datasets.

$$\text{Precision} = \frac{TP}{TP + FP}, \quad (2)$$

(iii) Recall is the proportion of actual positives that are correctly identified by the model.

$$\text{Recall} = \frac{TP}{TP + FN}, \quad (3)$$

(iv) F1-Score is the harmonic mean of precision and recall, providing a balanced metric when both values are important.

$$F_1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}, \quad (4)$$

(v) Loss quantifies the prediction error during training and is minimized to improve model performance.

$$\mathcal{L} = -[y \log(\hat{y}) + (1 - y) \log(1 - \hat{y})], \quad (5)$$

IV. RESULTS AND DISCUSSION

A. Overall Model Comparison

To develop a high-accuracy model used for land cover classification of satellite imagery, several deep learning models were explored and applied in DeepEarthMY Version 2.0. The results of the experiment is shown below

TABLE IV
COMPREHENSIVE MODEL PERFORMANCE COMPARISON

Model	Accuracy	Loss	Precision	Recall	F1-Score
vMambaV2-Small	0.8453	0.3659	0.8043	0.8511	0.8270
ResNet50	0.8467	0.8403	0.7463	0.9806	0.8476
EfficientNetV2-B2	0.9100	0.6098	0.8393	0.9806	0.9045
AlexNet	0.7792	0.4691	0.7021	0.8544	0.7708
ViT	0.7876	0.4637	0.7257	0.8220	0.7709

EfficientNetV2-B2 with learning rate decay achieved the best overall performance, with the highest accuracy of 91.00% and F1-score of 0.9045, along with strong recall of 0.9806 and precision of 0.8393. Its moderate loss of 0.6098 indicates good convergence and generalization, supported by compound scaling and progressive learning. Compared to ViT and VMamba, it converged faster with lower resource usage. Overall, EfficientNetV2-B2 exhibits excellent generalization due to its compound scaling and progressive learning.

B. EfficientNet and EfficientNetV2 Variant Comparison

After overall model comparison, we compared the classification performance across different versions of the EfficientNet family, including EfficientNetV2 (B0, B1, B2, B3) and EfficientNet (B5, B7). This is done to determine which version offers the best predictive accuracy for land-cover classification. The results are shown in Table V.

From Table V, EfficientNetV2-B2 demonstrates the best performance, achieving the highest accuracy (0.9100), precision (0.8393), and F1-score (0.9045). This indicates a strong balance between minimizing false positives and false negatives, making it the most reliable for general classification tasks. While EfficientNetV2-B0 records the lowest

TABLE V
CLASSIFICATION PERFORMANCE OF EFFICIENTNET VARIANTS

Model	Accuracy	Loss	Precision	Recall	F1-Score
EfficientNetV2-B0	0.8861	0.3008	0.8016	0.9806	0.8821
EfficientNetV2-B1	0.8650	0.4623	0.7766	0.9676	0.8617
EfficientNetV2-B2	0.9100	0.6098	0.8393	0.9806	0.9045
EfficientNetV2-B3	0.9001	0.3455	0.8324	0.9644	0.8936
EfficientNet-B5	0.8579	0.6007	0.7640	0.9741	0.8563
EfficientNet-B7	0.9058	0.5985	0.8306	0.9838	0.9007



Fig. 1. Training and validation of EfficientNetV2-B2 over epochs.

loss (0.3008), and EfficientNet-B7 achieves the highest recall (0.9838), these models trade off either precision or F1-score.

During training, EfficientNetV2-B2 showed high training accuracy with low loss, and despite fluctuations in validation accuracy (0.85–0.92) and a spike in loss at epoch 70, it consistently outperformed other models. This stability and generalization make it the most suitable model for deployment. In conclusion, EfficientNetV2-B2 has a strong balance between performance and model complexity, which is why it's outperforming the other variants.

C. VMamba Variant Comparison

Moreover, a detailed comparison of VMamba variants is carried out to identify the most suitable configuration within the family.

TABLE VI
CLASSIFICATION PERFORMANCE OF VMAMBA VARIANTS

Model	Accuracy	Loss	Precision	Recall	F1-score
VMambaV2-Tiny	0.8158	0.4227	0.8049	0.5340	0.6420
VMambaV2-Small	0.8453	0.3659	0.8043	0.8511	0.8270
VMambaV2-Base	0.8158	0.4314	0.8296	0.7249	0.7737

As shown in Table VI, VMambaV2-Small delivers the most balanced performance among the three variants, with the highest accuracy (0.8453), F1-score (0.8270), and recall (0.8511), indicating strong generalization and robustness. VMambaV2-Base achieves the highest precision (0.8296) but at the cost of lower recall and F1-score, making it suitable for precision-critical applications. In contrast, VMambaV2-Tiny has the weakest overall performance, particularly in recall (0.5340) and F1-score (0.6420), suggesting it may miss many positive cases. Training curves for VMambaV2-Small also show strong performance, although with minor fluctuations in validation

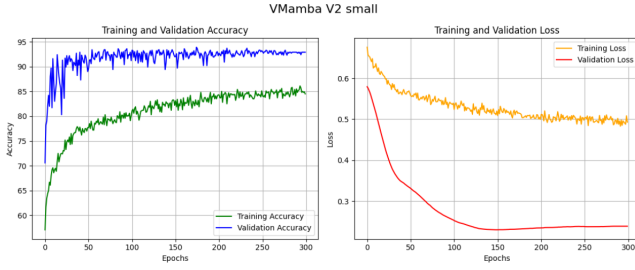


Fig. 2. Training and validation of VMambaV2-Small over epochs.

accuracy and slight overfitting tendencies after epoch 125. Overall, VMambaV2-Small strikes the best balance between complexity and performance for land-cover classification.

D. DeepEarthMY Version 1.0 and 2.0 Performance Comparison

After identifying EfficientNetV2-B2 as the best-performing model, it was further evaluated on both versions of the dataset to assess the impact of the additional annotations in DeepEarthMY V2.0. The experiment, using random and manual splits, aimed to highlight how the updated dataset and split strategy affect classification performance.

TABLE VII
COMPARISON OF MODEL PERFORMANCE ON DIFFERENT DATASET VERSIONS

Dataset Version	Accuracy	Loss	Precision	Recall	F1-Score
V1 (Random Split)	0.9526	0.1703	0.9146	0.9104	0.9278
V2 (Random Split)	0.9526	0.2499	0.9396	0.9492	0.9444
V2 (Manual Split)	0.9100	0.6098	0.8393	0.9806	0.9045

Table VII shows that the manually split V2 dataset yields lower accuracy and higher loss compared to the randomly split versions, despite a high recall. This suggests potential issues such as class imbalance or challenging samples in the test set. Random splits of V1 and V2 achieved better overall performance, with V2 (random) showing the best precision and F1 score. These findings indicate that the manual split may not be optimal and should be refined for better generalization.

V. CONCLUSION AND FUTURE WORK

This study investigated the effectiveness of several land-cover classification models. Additionally, a comparison within both VMamba and EfficientNet families were conducted to discover the best model in their respective groups. Moreover, further experiments on DeepEarthMY V1 and V2 datasets were conducted to evaluate the manual splitting strategy. Future work could focus on improving dataset quality with

additional geographically diverse imagery and applying advanced augmentation or semi-supervised learning to address class imbalance and enhance model generalization.

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